

Commutative diagrams with amscd

The square commutes when $h\alpha = \beta f$ and $k\beta = \gamma g$:

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ \alpha \downarrow & & \beta \downarrow & & \downarrow \gamma \\ D & \xrightarrow{h} & E & \xrightarrow{k} & F \end{array}$$

The same arrow grid includes equality, a dashed label, and opposing vertical arrows:

$$\begin{array}{ccccc} X & \xlongequal{\quad} & Y & \xrightarrow{u} & Z \\ \uparrow & & s \downarrow & & \downarrow t \\ P & \xleftarrow{v} & Q & \xleftarrow{w} & R \end{array}$$